WORKSHEET 1.8

The Three-Part Continuity Test

Unit 1: Limits and Continuity

Key ideas

- f is continuous at c when $f(c)$ is defined, $\lim_{x\to c} f(x)$ exists, and the two are equal.
- For a piecewise function the justification is one line: $\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = f(c)$.
- Polynomial, rational, root, trigonometric, exponential and logarithmic functions are continuous on their domains.
- To make pieces join, set the limits equal at each boundary: one equation per boundary.

Worked examples

Example 1 · Verifying continuity

Is $f(x) = \begin{cases} x^3, & x \le 1 \\ 3x - 2, & x > 1 \end{cases}$ continuous at $x = 1$?

Solution. $\lim_{x\to 1^-} f(x) = 1$, $\lim_{x\to 1^+} f(x) = 3 - 2 = 1$, and $f(1) = 1$.

Yes: the three values are equal.

Example 2 · One constant

Find k so that $f(x) = \begin{cases} kx - 3, & x < 2 \\ x^2 + k, & x \ge 2 \end{cases}$ is continuous at $x = 2$.

Solution. $2k - 3 = 4 + k$.

$k = 7$. **Check: both sides give 11.**

Example 3 · A function value and two constants

Find a and b so that $f(x) = \begin{cases} ax + 2, & x < 1 \\ 5, & x = 1 \\ bx^2, & x > 1 \end{cases}$ is continuous at $x = 1$.

Solution. Left limit: $a + 2$. Right limit: b. Both must equal $f(1) = 5$.

$a = 3$ **and** $b = 5$.

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Practice

CORE

1. Is $f(x) = \begin{cases} 4 - x, & x < 3 \\ x - 2, & x \geq 3 \end{cases}$ continuous at $x = 3$? Justify.

2. On what set is $f(x) = \dfrac{1}{x^2 - 9}$ continuous?

3. On what interval is $f(x) = \sqrt{x - 2}$ continuous?

4. What value should be assigned to $f(6)$ to make $f(x) = \dfrac{x^2 - 36}{x - 6}$ continuous at $x = 6$?

EXAM LEVEL

5. Find k so that $f(x) = \begin{cases} x^2 - k, & x \leq 2 \\ kx + 1, & x > 2 \end{cases}$ is continuous at $x = 2$.

6. Find k so that $f(x) = \begin{cases} \dfrac{\sin(4x)}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$.

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7. Find a and b so that $f(x) = \begin{cases} x + a, & x < 0 \\ x^2 + 3, & 0 \le x \le 2 \\ bx - 1, & x > 2 \end{cases}$ is continuous for all real x.

8. On what interval is $f(x) = \ln(5 - x)$ continuous?

Question 9 · Free-response practice

Let $f(x) = \begin{cases} x^2 + 1, & x < 2 \\ 5, & x = 2 \\ 7 - x, & x > 2. \end{cases}$

(a) Find $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$.

(b) Is f continuous at $x = 2$? Justify your answer.

(c) Let $h(x) = f(x)$ for $x \neq 2$ and $h(2) = 4$. Is h continuous at $x = 2$? If not, name the type of discontinuity.

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Worked solutions

1. $\lim\limits_{x\to 3^-} f(x) = 1$, $\lim\limits_{x\to 3^+} f(x) = 1$, $f(3) = 1$. All equal, so yes.
2. All real numbers except $x = 3$ and $x = -3$.
3. $[2, \infty)$.
4. 12, the limit of $x + 6$.
5. $4 - k = 2k + 1$ gives $k = 1$.
6. $\lim\limits_{x\to 0} \dfrac{\sin 4x}{x} = 4$, so $k = 4$.
7. At 0: $a = 3$. At 2: $7 = 2b - 1$, so $b = 4$.
8. $5 - x > 0$: the interval $(-\infty, 5)$.
9. (a) $\lim\limits_{x\to 2^-} f(x) = 4 + 1 = 5$ and $\lim\limits_{x\to 2^+} f(x) = 7 - 2 = 5$.
   (b) Yes. $\lim\limits_{x\to 2^-} f(x) = \lim\limits_{x\to 2^+} f(x) = f(2) = 5$.
   (c) No. $\lim\limits_{x\to 2} h(x) = 5$ but $h(2) = 4$: a removable discontinuity.

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